

6. It is well known that plastic bags are a major source of litter. These bags can exist for a very long time, ranging from tens of years in the natural environment up to a thousand years in landfill.

The use of plastic bags is now banned in some countries. These countries commonly use bags made from paper or cotton, which do not cause the same litter problem as plastic bags.

Paper or plastic?



Most people assume that a paper bag is better for the environment, however because paper bags are almost ten times heavier than plastic bags, they produce a greater mass of waste. On the other hand, because paper bags are biodegradable they do not lead to the same problem with litter as plastic bags.

When deciding whether paper is more suitable than plastic for making shopping bags, information from life cycle assessments (LCAs) of both materials must be taken into consideration.

Some of the findings from LCAs of paper and plastic bags are shown below.

Plastic bags are toxic to aquatic environments and their production is more likely to deplete natural resources

Plastic bags use 70 % less energy in their production compared to paper bags

The processes used in paper production cause atmospheric acidification and eutrophication

The production of 1 million paper bags produces 51 tonnes of carbon dioxide, compared to 31 tonnes for plastic bags

693 kJ of energy is needed to recycle 1 kg of paper bags compared to 8 kJ for the same mass of plastic bags

4 546 litres of water are used to produce 1 000 paper bags compared to 177 litres for the same number of plastic bags

Paper bags are made from trees and plastic bags are made from ethene

Paper bags are 10 times heavier than plastic bags and therefore have more associated costs

On reading these findings it is important to consider all factors relating to sustainability when deciding on the most suitable material to make shopping bags. We should not just be thinking about what happens when we throw them away!



Examiner
only

- (a) Use the information in the passage to compare the mass of waste and volume of litter generated from the use of plastic and paper bags. [2]

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- (b) Use the information from the LCAs to tick (✓) whether each of the factors supports or opposes the use of paper bags in preference to plastic bags. [3]

	Supports use of paper bags	Opposes use of paper bags
Impact of waste on marine life	☐	☐
Water consumption in production	☐	☐
Energy used in production	☐	☐
Carbon footprint generated in production	☐	☐
Energy used in recycling	☐	☐
Cost of transporting waste	☐	☐

- (c) The use of which type of bag is more likely to be linked to the erosion of limestone buildings?
Give a reason for your answer. [1]

Type of bag

Reason

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